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C20-C–**501**

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BOARD DIPLOMA EXAMINATION, (C-20)

OCTOBER/NOVEMBER—2024

DCE – FIFTH SEMESTER EXAMINATION

STEEL STRUCTURES

Time : 3 Hours] [*Total Marks* : 80

PART—A

3×10=30

Instructions : (1) Answer **all** questions.

(2) Each question carries **three** marks.

(3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

(4) Use of IS-800 : 2007 and Steel Table are allowed.

1. State the load combinations considered in design of steel structures.

2. Write the provisions for maximum size of fillet weld as per IS : 800-2007.

3. Calculate the safe load transmitted by field fillet welded joint, if size of weld is 8 mm and length 300 mm. Take f_u = 410 N/mm^2.

4. Define the terms *(a)* gross area, *(b)* net area and *(c)* net effective area of a tension member.

5. State three factors affecting the strength of a compression member.

6. Sketch the sectional elevation of slab base and label the parts.

7. Write any three codal provisions to be followed in the design of lacing system as per IS : 800-2007.

8. Define the terms *(a)* elastic moment of resistance, *(b)* plastic moment of resistance and *(c)* shape factor.

9. Draw the cross-section of a plate girder and mention the component parts.

10. List any six types of roof trusses commonly used for different spans.

PART—B

8×5=40

Instructions : (1) Answer either *(a)* or *(b)* from each questions from Part-B.

(2) Each question carries **eight** marks.

(3) Answers should be comprehensive and criterion for valuation is the content but not the length of the answer.

11. *(a)* A double angle tension member 2 ISA 110 mm × 110 mm × 12 mm carrying an axial tension of 700 kN is to be connected to a 12 mm thick gusset plate by side and end welds. Design the joint if the ultimate shear stress in the weld is 270 N/mm^2. Assume connections are made in site.

(OR)

(b) Design a lap joint for two plates 200 mm × 6 mm and 200 mm × 8 mm using maximum permissible size of fillet welds. The ultimate shear stress in the fillet weld is 410 MN/m^2. Assume that the connections are made in workshop. Also determine maximum permissible tensile stress in the thinner plate.

12. *(a)* Design a single unequal angle tension member of a roof truss to carry factored tensile force of 275 kN. The long leg of the angle is to be connected to a gusset plate by fillet welds. $f_y = 250$ N/mm^2, $f_u = 410$ N/mm^2. Assume length of weld as 150 mm.

(OR)

(b) Determine the design strength of a tensile member ISA 100 mm × 75 mm × 10 mm when its longer leg is connected to 10 mm gusset plate by 6 mm size fillet welds. The effective length of the weld is 170 mm. Take $f_y = 250$ N/mm^2 and $f_u = 410$ N/mm^2.

13. *(a)* Determine the design compressive strength of single ISHB 400 @ 774 N/m, when it is used as column of 5·5 m height with both ends fixed. $f_y = 250$ N/mm^2.

(OR)

(b) Design a single unequal angle discontinuous strut to carry a compressive force of 100 kN. The centre to centre distance between the end connections is 2 m. Assume that the end connections are made with fillet welds and gusset fixity is hinged. $f_y = 250$ N/mm^2 and $E = 2 \times 10^5$ N/mm^2.

14. *(a)* Design a slab base with rectangular base plate having equal projections for a column section consisting of ISHB 250 @ 536 N/m carrying an axial load of 1000 kN including self-weight. Use M-20 grade concrete and Fe-250 grade steel. Also design the concrete pedestal if safe bearing capacity of soil is 180 kN/m^2.

(OR)

(b) Determine the design compressive strength of single angle discontinuous strut ISA 125 mm × 95 mm × 10 mm of length 2 m. The longer leg is connected to the gusset plate by fillet weld and gusset fixity is hinged. Take f_y = 300 N/mm^2 and $E = 2 \times 10^5$ N/mm^2.

15. *(a)* A simply supported laterally restrained beam ISMB 400 @ 616 N/m is subjected to bending moment of 100 kNm and shear force of 80 kN. Check the safety of the beam in bending and shear. The yield stress of steel is 250 N/mm^2.

(OR)

(b) Design a simply supported beam of an effective span 5 m and carries a udl of 25 kN/m including self-weight. The compression flange of beam is laterally restrained. Check the beam for shear and deflection the grade of steel is Fe-250.

PART—C

10×1=10

Instructions : (1) This question is compulsory.

(2) The question carries **ten** marks.

(3) Answer should be comprehensive and the criterion for valuation is the content but not the length of the answer.

16. A discontinuous double angle strut consists of 2 ISA 90 mm × 60 mm × 10 mm are connected to both sides of gusset plate of 12 mm thick at both ends by suitable fillet welds. The centre to centre distance of end connections are 2·5 m. The yield stress of steel is 340 N/mm^2. Determine the design compressive force that can be carried by the strut. The gusset fixity may be taken as hinged.

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